Procurement & commissioning plan

SO3 Converter Ver 1 · SO3-V1-PC-CMS-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKED / APPROVED
0	27 Oct 2026	Issued for teaching	Apinan Soottitantawat	—

Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose, scope and basis

This plan takes the converter section of SO3 Converter Ver 1 from the end of design to handover: the EPC sequence, the procurement packages with their long-lead items and inspection levels, a schematic schedule with its critical path, FAT and SAT (including one SAT step per SIS trip of the C&E), the mechanical-completion and pre-commissioning checklist per system, commissioning mapped to the operator-training procedures, and the performance test.

Battery limits as the simulator model: dried air in, molten sulfur in, acid and steam out, tail gas to the stack. Design point (model): sulfur 40.94 t/h, air 238.5 kNm³/h, SO_2 into bed 1 12.0 %, conversion 99.80 %, stack 292.5 ppm, acid 125.0 t/h, steam 94.0 t/h, blower 4200 kW.

Durations are typical practice and are labelled so; they are assumptions for teaching, not a contract schedule. Related documents: SO3-PID-001/002 (EPC P&IDs), SO3-V1-IN-CE-001 (C&E), SO3-V1-IN-SRS-001 (SRS), SO3-V1-SA-HZP-001 (HAZOP report), PFD-201, SO3-V1-PE-HMB-001, SO3-V1-ME-EQL-001, SO3-V1-IN-INL-001, SO3-V1-EE-LL-001.

2 · EPC sequence after design

#	Stage	Who does what	Output	Typical duration	Note
1	Material requisition (MR)	Engineering (discipline lead) → Procurement	MR with datasheets, specifications, VDRL, inspection level	2–4 weeks per package (typical)	Issued from the approved datasheet (EQL, INL, HMB). Long-lead MRs are issued first, from the process design freeze.
2	Enquiry and bid (RFQ)	Procurement; bidders	Bid package to 3–5 qualified bidders; clarification log	4–6 weeks bid period (typical)	Commercial and technical bids received separately.
3	Technical bid evaluation (TBE)	Engineering (lead), Process, Inspection; Owner comments	TBE sheet: compliance matrix, deviations, technical ranking	2–4 weeks (typical)	Only technically acceptable bids go to the commercial evaluation.
4	Purchase order (PO)	Procurement; Owner approval above a value limit	PO with the final datasheet, VDRL, inspection and test plan (ITP)	1–2 weeks after the recommendation (typical)	The lead time is counted from the PO.
5	Vendor documents	Vendor submits; Engineering reviews (codes 1–4)	GA, P&ID of the package, calculations, ITP, data book	first issue 4–8 weeks after PO; review 2 weeks per cycle (typical)	Certified vendor data (nozzle loads, weights, I/O) feed piping, civil and the DCS.
6	Inspection and FAT	Vendor; Owner/contractor inspectors witness hold points	Inspection release note, FAT report, punch list	from the PO to shipment = the lead time; FAT 1–10 days (typical)	Inspection level of the package sets which hold points are witnessed.
7	Delivery and receiving	Logistics; site material control	Receiving report, preservation record	2–8 weeks transport (typical)	Preservation until erection: dry air or N_2 on exchangers and the blower, catalyst kept dry.
8	Construction	Construction contractor; QA/QC	Installation records, weld and NDT records, test packs	12–18 months on site (typical)	Civil → setting → steel → piping and ducting → E&I.
9	Mechanical completion (MC)	Construction hands over to the commissioning team, system by system	MC certificate per system with an A/B punch list	per system (typical)	A-punch items must be closed before pre-commissioning; B items may follow.
10	Pre-commissioning	Commissioning team with vendors	Cleaned, tested, dried systems; loop checks; SIS proof test	1–3 months (typical)	No process fluid yet (air, water, weak acid, steam).
11	Commissioning	Commissioning team with Operations	Systems run on their real fluids; SAT closed; PSSR	2–6 weeks (typical)	Acid circulation, molten sulfur on recirculation, steam system, catalyst preheat.
12	Start-up	Operations (commissioning team supporting)	First sulfur through the SIS; line-out	model cold start-up: sulfur admitted after 194 min, lined out at 274 min	Trained on the OTS: procedure "Cold start-up through the SIS".
13	Performance test	Operations, Process, contractor; Owner witnesses	Performance test report against the guarantees	72 h continuous (typical)	After a stable run at design rate.
14	Handover	Contractor → Owner	Handover certificate, as-built documents, spares, data books	—	Care, custody and control move to the Owner.

3 · Procurement packages

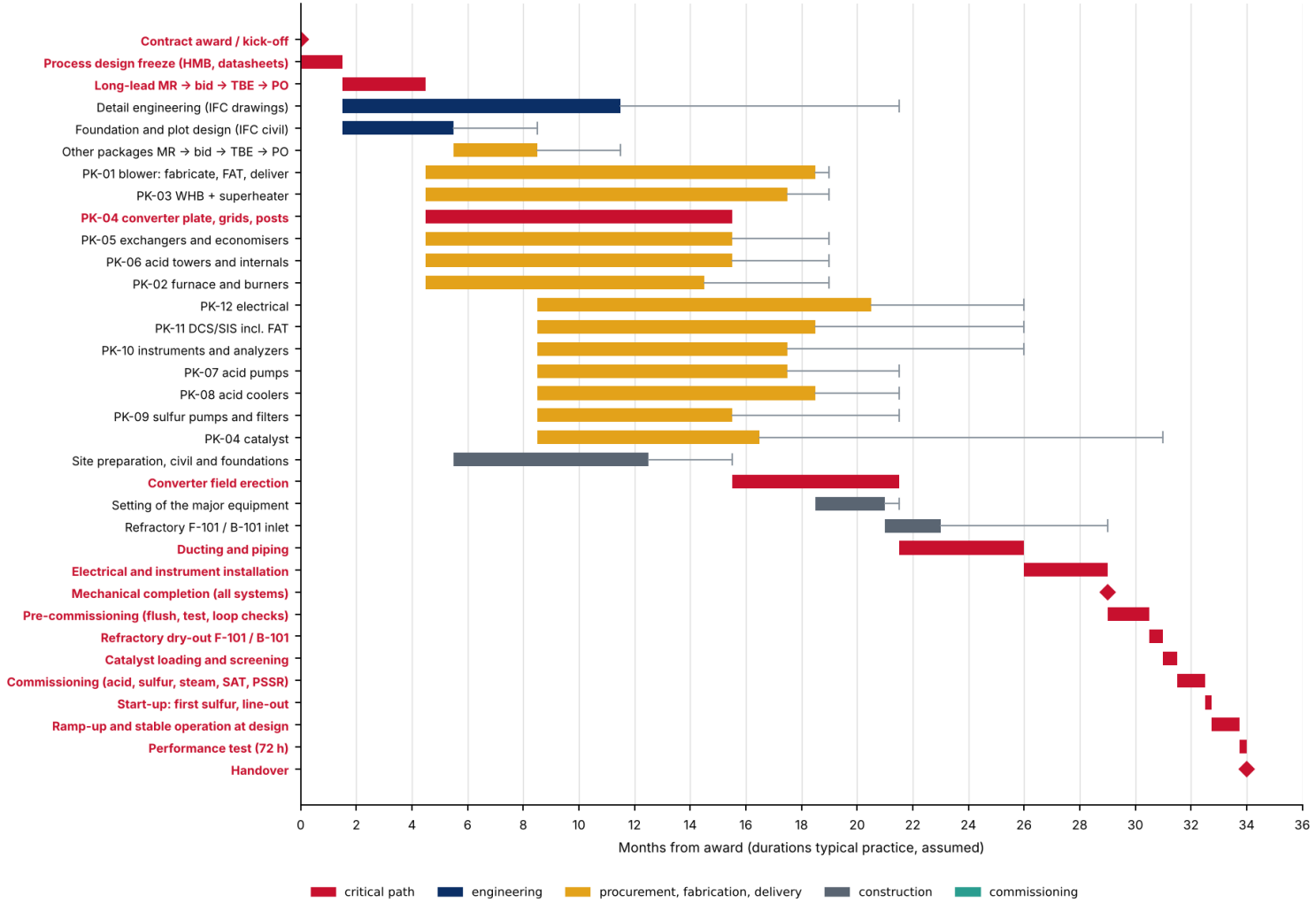
A package is long-lead when its lead time from the PO is 9 months or more (typical threshold). Inspection levels follow a typical four-level scheme (assumed for this unit). The critical-path flag comes from the schedule network in section 4.

Level	Name	What the inspector does (typical)
1	Level 1 · full	Pre-inspection meeting, resident or frequent visits, all hold points witnessed, FAT witnessed
2	Level 2 · key points	Material certificates reviewed, hydro/performance test and final inspection witnessed
3	Level 3 · final	Document review and final visual/dimensional inspection before shipment
4	Level 4 · certificates	Certificates of conformity reviewed on receipt only

Package	MR	Equipment	Scope (model numbers)	Lead time (typical)	Insp.	Shop test / FAT	Critical path / float
PK-01 Main air blower and drive	MR-01	K-101	Centrifugal blower 238.5 kNm³/h dried air at 90 °C, discharge PI-101 60 kPa g; ≈4.2 MW motor; lube-oil system, inlet filter, silencer	14 months · long-lead	1	Mechanical run test 4 h, performance test on the curve, vibration and bearing temperatures, motor no-load test	float 0.5 months
PK-02 Sulfur furnace and burners	MR-02	F-101, H-101	Refractory-lined furnace at 1121 °C, 14 % SO_2; sulfur guns; H-101 start-up preheat burner with its flame safeguard	10 months · long-lead	2	Burner management FAT (flame safeguard, low-air interlock), shell hydro/leak test, refractory material certificates	float 4.5 months
PK-03 Waste heat boiler and superheater	MR-03	B-101, SH-201	Fire-tube boiler, steam 94 t/h at 4.0 MPa g, gas 1121 → 274 °C; hot-gas bypass TV-201; superheater SH-201 (2.3 MW)	13 months · long-lead	1	Hydrotest to the pressure code, tube-to-tubesheet weld NDT, ferrule fit-up, final inspection	float 1.5 months
PK-04 Converter and catalyst	MR-04	R-201	Five-bed converter, inside diameter 13.0 m (assumed, Design tab), grids and posts; catalyst 187.1 t / 340 m³: V_2O_5–K_2SO_4 beds 1–3 116.1 t, Cs-promoted beds 4–5 71.0 t	11 months · long-lead	1	Shell plate and weld inspection, grid load test; catalyst: certificate of analysis, sample screening loss, bulk density	critical
PK-05 Gas–gas exchangers and economisers	MR-05	E-201, E-202, E-203, E-204, E-205	E-201 hot interpass 17.2 MW, E-202 cold interpass 9.2 MW; economisers E-203 27.9 MW, E-204 1.0 MW, E-205 20.8 MW (BFW 150 t/h); bypasses TV-211…TV-251	11 months · long-lead	2	Shell and tube-side pressure/leak test, tube expansion and weld NDT, dimensional check of the duct nozzles	float 3.5 months
PK-06 Acid towers, distributors and mist eliminators	MR-06	T-301, T-302, TK-301, TK-302	IPA T-301 (1600 m³/h 98.5 % acid) and FAT T-302 (1000 m³/h), brick-lined, packing, acid distributors, candle mist eliminators (stack mist design 10.5 mg/Nm³); pump tanks TK-301/302	11 months · long-lead	2	Distributor water-flow test (uniformity), mist eliminator candle ΔP and efficiency certificate, shell leak test	float 3.5 months
PK-07 Acid circulation pumps	MR-07	P-301A, P-301B, P-302A, P-302B	Vertical submerged acid pumps: P-301A/B 1600 m³/h, P-302A/B 1000 m³/h, alloy wetted parts; HS-301B auto-start	9 months · long-lead	2	Hydraulic performance test (head–flow, power), mechanical run, NPSH check (typical pump standard)	float 4.0 months
PK-08 Acid coolers	MR-08	no tag issued	Coolers on the IPA and FAT acid loops (outside the model; shown as "acid cooler + dilution (note 2)" on SO3-PID-002, no tag issued); anodic protection typical	10 months · long-lead	2	Pressure test both sides, anodic-protection system function test	float 3.0 months
PK-09 Sulfur pumps and filters	MR-09	P-101A, P-101B	Steam-jacketed molten-sulfur pumps P-101A/B for 40.94 t/h at ≈140 °C; steam-jacketed sulfur filters (PDG-103, no equipment tag issued)	7 months	3	Pump performance test on water, jacket pressure test, filter element certificate	float 6.0 months
PK-10 Instruments, valves and analyzers	MR-10	FV-101, FV-102, FV-103, XV-103, TV-201, TV-211, TV-221, TV-231, TV-241, TV-251, HV-112	Control valves FV-101/102/103 and TV-201…251, SIS valve XV-103 (SIL-certified), HV-112; flow, pressure and temperature transmitters, bed thermocouples; analyzers AI-104 (SO_2), AI-105 (O_2), AIT-306 stack SO_2, AIT-307 acid mist	9 months · long-lead	2	Valve seat leakage and stroke tests, fail-action check, analyzer FAT with test gas, calibration certificates	float 8.5 months
PK-11 DCS and SIS	MR-11	no tag issued	DCS: 12 control loops, 21 alarms; SIS: 6 trips (I-1…I-6), 5 SIFs, one final element XV-103; HMI, historian	10 months · long-lead	1	Integrated FAT: I/O, every C&E row, loop configuration, graphics, alarm list, response time, bypass and reset logic	float 7.5 months
PK-12 Electrical	MR-12	no tag issued	HV switchgear and the K-101 motor feeder (≈4.2 MW), transformers, MCCs for the pumps, UPS for DCS/SIS, cables, earthing, lighting	12 months · long-lead	2	Switchgear routine tests, transformer routine tests, protection relay settings, UPS autonomy test	float 5.5 months

4 · Schedule from award to performance test

Schematic network of 34 activities, finish-to-start links, durations typical practice (assumed). Award to handover 34.0 months; typical for a sulfur-burning acid plant of this size 28–36 months (typical practice).



Critical path: Process design freeze (HMB, datasheets) → Long-lead MR → bid → TBE → PO → PK-04 converter plate, grids, posts → Converter field erection → Ducting and piping → Electrical and instrument installation → Pre-commissioning (flush, test, loop checks) → Refractory dry-out F-101 / B-101 → Catalyst loading and screening → Commissioning (acid, sulfur, steam, SAT, PSSR) → Start-up: first sulfur, line-out → Ramp-up and stable operation at design → Performance test (72 h).

ID	Activity	Predecessors	Duration (months)	Early start	Early finish	Total float	Critical
K0	Contract award / kick-off	—	0.00	0.00	0.00	0.00	yes
BED	Process design freeze (HMB, datasheets)	K0	1.50	0.00	1.50	0.00	yes
LLP	Long-lead MR → bid → TBE → PO	BED	3.00	1.50	4.50	0.00	yes
ENG	Detail engineering (IFC drawings)	BED	10.00	1.50	11.50	10.00	
ENGF	Foundation and plot design (IFC civil)	BED	4.00	1.50	5.50	3.00	
BLK	Other packages MR → bid → TBE → PO	ENGF	3.00	5.50	8.50	3.00	
F_K101	PK-01 blower: fabricate, FAT, deliver	LLP	14.00	4.50	18.50	0.50	
F_WHB	PK-03 WHB + superheater	LLP	13.00	4.50	17.50	1.50	
F_CONV	PK-04 converter plate, grids, posts	LLP	11.00	4.50	15.50	0.00	yes
F_GGE	PK-05 exchangers and economisers	LLP	11.00	4.50	15.50	3.50	
F_TWR	PK-06 acid towers and internals	LLP	11.00	4.50	15.50	3.50	
F_FUR	PK-02 furnace and burners	LLP	10.00	4.50	14.50	4.50	
F_EL	PK-12 electrical	BLK	12.00	8.50	20.50	5.50	
F_DCS	PK-11 DCS/SIS incl. FAT	BLK	10.00	8.50	18.50	7.50	
F_INS	PK-10 instruments and analyzers	BLK	9.00	8.50	17.50	8.50	
F_PMP	PK-07 acid pumps	BLK	9.00	8.50	17.50	4.00	
F_ACC	PK-08 acid coolers	BLK	10.00	8.50	18.50	3.00	
F_SUL	PK-09 sulfur pumps and filters	BLK	7.00	8.50	15.50	6.00	
F_CAT	PK-04 catalyst	BLK	8.00	8.50	16.50	14.50	
CIV	Site preparation, civil and foundations	ENGF	7.00	5.50	12.50	3.00	
ERC	Converter field erection	F_CONV, CIV	6.00	15.50	21.50	0.00	yes
SET	Setting of the major equipment	F_K101, F_WHB, F_GGE, F_TWR, F_FUR, CIV	2.50	18.50	21.00	0.50	
REF	Refractory F-101 / B-101 inlet	SET	2.00	21.00	23.00	6.00	
PIP	Ducting and piping	SET, ERC, ENG, F_PMP, F_ACC, F_SUL	4.50	21.50	26.00	0.00	yes
EI	Electrical and instrument installation	PIP, F_EL, F_DCS, F_INS	3.00	26.00	29.00	0.00	yes
MC	Mechanical completion (all systems)	EI, REF	0.00	29.00	29.00	0.00	yes
PRE	Pre-commissioning (flush, test, loop checks)	MC	1.50	29.00	30.50	0.00	yes
DRY	Refractory dry-out F-101 / B-101	PRE	0.50	30.50	31.00	0.00	yes
CATL	Catalyst loading and screening	DRY, F_CAT	0.50	31.00	31.50	0.00	yes
COM	Commissioning (acid, sulfur, steam, SAT, PSSR)	CATL	1.00	31.50	32.50	0.00	yes
SU	Start-up: first sulfur, line-out	COM	0.25	32.50	32.75	0.00	yes
RAMP	Ramp-up and stable operation at design	SU	1.00	32.75	33.75	0.00	yes
PT	Performance test (72 h)	RAMP	0.25	33.75	34.00	0.00	yes
HO	Handover	PT	0.00	34.00	34.00	0.00	yes

5 · FAT and SAT

5.1 SIS · SAT steps generated from the C&E (SO3-V1-IN-CE-001)

Each trip row of the C&E becomes a SAT step: inject the cause, check every effect of the row, then hold the cause for less than its delay and check that nothing trips. Steps SAT-07 and SAT-08 test the two sequence actions that are not SIS functions.

Step	Cause (C&E)	Set point	Delay	Test method	Expected effects	Result / sign
SAT-01	I-6 ESD — Emergency shutdown push-button (SIS screen)	—	0 s	Press the ESD push-button on the SIS screen with sulfur admitted on the test header	XV-103 closes; FIC-103 forced to MAN 0 % (FV-103 closes); AIC-104 to MAN; K-101 stops; H-101 shut off; SIS TRIPPED and latched; first-out shown; trip horn and annunciation	
SAT-02	FSLL-100 — Total air low-low trip, I-1 (FT-100)	90 kNm³/h	3 s	Inject a signal at FT-100 below the set point (transmitter in test, XV-103 open on the isolated test header); repeat for a time shorter than the delay — no trip	XV-103 closes; FIC-103 forced to MAN 0 % (FV-103 closes); AIC-104 to MAN; SIS TRIPPED and latched; first-out shown; trip horn and annunciation	
SAT-03	TSHH-203 — Bed 1 exit high-high trip, I-2 (TE-203)	660 °C	5 s	Inject a millivolt signal at TE-203 above the set point; check that the TI-203 indication and the 640 °C alarm see the same signal; check the trip blocks the SIS reset while high	XV-103 closes; FIC-103 forced to MAN 0 % (FV-103 closes); AIC-104 to MAN; SIS TRIPPED and latched; first-out shown; trip horn and annunciation	
SAT-04	FSLL-301 — IPA acid circulation low-low trip, I-3 (FT-301)	800 m³/h	10 s	Reduce the FT-301 signal below the set point; check first that the P-301B auto-start acts at the higher point (one shot), then the trip; repeat shorter than the delay — no trip	XV-103 closes; FIC-103 forced to MAN 0 % (FV-103 closes); AIC-104 to MAN; SIS TRIPPED and latched; first-out shown; trip horn and annunciation	
SAT-05	FSLL-302 — FAT acid circulation low-low trip, I-4 (FT-302)	500 m³/h	10 s	Reduce the FT-302 signal below the set point; repeat shorter than the delay — no trip	XV-103 closes; FIC-103 forced to MAN 0 % (FV-103 closes); AIC-104 to MAN; SIS TRIPPED and latched; first-out shown; trip horn and annunciation	
SAT-06	ASHH-306 — Stack SO_2 high-high trip, I-5 (AIT-306)	3000 ppm	300 s	Apply test gas or a simulated AIT-306 signal above the set point: no trip during the 60 min inhibit after sulfur admission; after the inhibit, trip after the 5 min delay	XV-103 closes; FIC-103 forced to MAN 0 % (FV-103 closes); AIC-104 to MAN; SIS TRIPPED and latched; first-out shown; trip horn and annunciation	
SAT-07	FI-100 — H-101 preheat burner low-air interlock (burner protection, not SIS)	60 kNm³/h	0 s	Reduce the FI-100 signal below 60 kNm³/h	H-101 shut off; DCS alarm	
SAT-08	FI-301 — P-301B auto-start on low IPA acid flow (HS-301B armed; one shot)	960 m³/h	3 s	Reduce the FI-301 signal below 960 m³/h for longer than 3 s	P-301B starts; DCS alarm	

5.2 SIS · start permissives

Step	Permissive (model purgePermissives)	Test	Result / sign
SAT-P1	SIS reset (IDLE), no trip active	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P2	K-101 running, total air FI-100 ≥ 100 kNm³/h	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P3	Furnace F-101 hot enough to ignite sulfur: TI-106 ≥ 450 °C	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P4	Bed 1 above strike temperature: TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P5	Bed 4 catalyst above strike temperature: TI-232 ≥ 360 °C	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P6	Acid circulating in T-301 and T-302 (FI-301, FI-302 ≥ 80 %)	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P7	Start-up line HV-112 to bed 4 closed	With all others met, make this one false: the SIS refuses to open XV-103 and names it	
SAT-P8	FIC-103 in MAN with output ≤ 25 %	With all others met, make this one false: the SIS refuses to open XV-103 and names it	

5.3 SIS · FAT

- Hardware and I/O: every SIS input and output wired to its terminal, channel tested with a simulator
- Logic: every C&E row of SO3-V1-IN-CE-001 tested cause by cause, with its delay, latch and first-out

- Start permissives: each one blocks the sulfur admission on its own (list below)
- Response time: sensor 0.5 s + logic 0.25 s + valve stroke 2 s (assumed, SRS) checked against ½ process safety time
- Reset: blocked while TSHH-203 is active; FSLL-100/301/302 and ASHH-306 do not block it
- Bypass and override: maintenance bypass only with a permit key, alarm and timer; start-up inhibit of ASHH-306 by logic only
- Fail-safe: loss of power or communication de-energises the outputs (XV-103 closes)
- Communication to the DCS: SIS state, first-out, trip causes and the inhibit timer shown on the DCS

5.4 DCS · FAT and SAT

- I/O check against the instrument list (channel, range, engineering units)
- Every control loop: mode changes MAN/AUTO/CAS, bumpless transfer, direct/reverse action, output limits
- Ratio station FFC-101 (air/sulfur) and the AIC-104 → FIC-102 cascade; TIC-201 and TIC-211…251 on their bypass valves
- Valve fail positions configured as built: FV-101/102 open, FV-103 closed, TV-201 closed, TV-211…251 open
- Alarm list: every alarm with its limit, priority and suppression while sulfur is off
- Graphics: every tag on the P&ID shown, trends, the SIS overview
- SAT: loop checks from the field transmitter to the HMI and from the HMI to the valve (stroke 0–25–50–75–100 %)

5.5 K-101 main air blower · FAT and site run

- Shop: mechanical run test (4 h typical) at rated speed; vibration and bearing temperature within the vendor limits
- Shop: performance test at the duty point 238.5 kNm³/h, 60 kPa g (curve with ±5 % tolerance band, typical)
- Shop: motor no-load and heat-run tests; driver ≈4.2 MW at design (model: blower power 4200 kW)
- Site: lube-oil flushing to cleanliness class, alignment, motor solo run 2 h (typical)
- Site: coupled run on air through the dried-air system and the open converter train; FI-100 calibrated against the blower curve
- Site: trip tests of the blower protections; check that K-101 stop is seen by FSLL-100 and the H-101 low-air interlock

6 · Mechanical completion and pre-commissioning checklist

Handover from construction is by system. MC items prove the system is built as designed; PC (pre-commissioning) items make it clean, tight, dry and checked before any process fluid. All checks are typical practice.

6.1 Sulfur

Done	Subsystem	Type	Check (typical practice)	Date / sign
☐	SUL-1 Sulfur pumps and filters P-101A/B	MC	Pumps set, aligned, jacket steam connected; filters assembled with elements	
☐	SUL-1 Sulfur pumps and filters P-101A/B	PC	Steam-jacket and tracing pressure test; tracing heats lines to ≈140 °C before sulfur	
☐	SUL-1 Sulfur pumps and filters P-101A/B	PC	Run-in on recirculation with molten sulfur (sulfur tank farm ready)	
☐	SUL-2 Sulfur line to F-101, FV-103, XV-103	PC	Line flushed and blown dry; hydro/pneumatic test (typical 1.5 × / 1.1 × design pressure)	
☐	SUL-2 Sulfur line to F-101, FV-103, XV-103	PC	XV-103 full-stroke and closure time; FV-103 fail-closed check	

6.2 Air and furnace

Done	Subsystem	Type	Check (typical practice)	Date / sign
☐	AIR-1 K-101 blower and air ducts	MC	Ducts cleaned and inspected; blower set, grouted, aligned	
☐	AIR-1 K-101 blower and air ducts	PC	Lube-oil flushing; motor solo run; coupled run on air	
☐	AIR-2 Air control FV-101, FV-102	PC	Stroke and fail-open check; FI-100 / FIC-101 / FIC-102 loop checks	
☐	AIR-3 Furnace F-101 and H-101	MC	Refractory installed and inspected; burner and sulfur guns fitted	
☐	AIR-3 Furnace F-101 and H-101	PC	Refractory dry-out of F-101 to the refractory supplier's heating curve (H-101 fired, blower air)	
☐	AIR-3 Furnace F-101 and H-101	PC	H-101 flame safeguard and low-air interlock (60 kNm³/h) tested	

6.3 WHB and steam

Done	Subsystem	Type	Check (typical practice)	Date / sign
☐	WHB-1 B-101 and steam drum	MC	Hydrotest records reviewed; inlet tubesheet refractory and ferrules installed	
☐	WHB-1 B-101 and steam drum	PC	Refractory dry-out of the B-101 inlet chamber together with F-101	
☐	WHB-1 B-101 and steam drum	PC	Chemical cleaning (alkaline boil-out typical) and flushing of the water side	
☐	WHB-2 Superheater SH-201, steam lines	PC	Steam blowing of the superheated-steam lines to target plates (typical)	
☐	WHB-3 Economiser BFW E-203/204/205	PC	BFW flushing; FI-253 loop check; safety valves set and sealed	
☐	WHB-4 Hot-gas bypass TV-201	PC	Stroke and fail-closed check; TIC-201 loop check	

6.4 Converter

Done	Subsystem	Type	Check (typical practice)	Date / sign
☐	CNV-1 R-201 shell and beds	MC	Internal inspection: grids, posts, division plates, manways; cleanliness	
☐	CNV-1 R-201 shell and beds	PC	Leak test of the converter and gas ducts with blower air (soap test of flanges, typical)	
☐	CNV-2 Catalyst loading	PC	Screen the catalyst to remove fines; load and level each bed (33.6 / 37.4 / 45.1 / 37.4 / 33.6 t); quartz-ring layers; record bed depths	
☐	CNV-2 Catalyst loading	PC	Close manways after the final inspection; record bed ΔP on air (PDI-204…244) as the clean baseline	
☐	CNV-3 Interpass exchangers and bypasses	PC	TV-211…TV-251 stroke and fail-open check; bed thermocouples TI-2x1/2x2/2x3 loop checked	
☐	CNV-4 Start-up line HV-112	PC	HV-112 operates, then is locked closed; lock recorded	

6.5 Absorption

Done	Subsystem	Type	Check (typical practice)	Date / sign
☐	ABS-1 Towers T-301 / T-302	MC	Brick lining, packing and distributors inspected; mist-eliminator candles installed and sealed	
☐	ABS-2 Acid loops P-301A/B, P-302A/B, coolers	PC	Circulate water first: leak check, distributor flow pattern, pump run-in	
☐	ABS-2 Acid loops P-301A/B, P-302A/B, coolers	PC	Drain and dry; then weak acid; then raise to 98.5 % acid strength before any SO_3 arrives	
☐	ABS-2 Acid loops P-301A/B, P-302A/B, coolers	PC	HS-301B auto-start of P-301B tested; FI-301 / FI-302 loop checks	
☐	ABS-3 Stack analyzers AIT-306 / AIT-307	PC	Analyzer calibration with test gas; data link to the DCS	

6.6 Utilities, DCS and SIS

Done	Subsystem	Type	Check (typical practice)	Date / sign
☐	UTL-1 Instrument air	PC	Header blown clean, dew point checked; air-failure test of every valve fail position	
☐	UTL-2 Electrical	PC	Insulation tests, protection relay settings, motor rotation checks, UPS autonomy	
☐	UTL-3 DCS	PC	Loop checks, alarm test, graphics against the P&ID	
☐	UTL-4 SIS	PC	SIS proof test of every SIF before first sulfur (SAT steps in part 4); permissives tested	
☐	UTL-5 Safety equipment	PC	Safety showers, acid-resistant PPE, SO_2/SO_3 gas detectors and the emergency plan in place	

7 · Commissioning to start-up, mapped to the OTS procedures

Each commissioning step names the procedure that trains it on the simulator: the graded cold start-up (steps P1–P14, listed in 7.2) and the SOP register (SOP-SO3-01 … 12). Model timing of the cold start-up: sulfur admitted after 194 min and lined out after 274 min from the blower start (Transition operation runs).

7.1 Commissioning steps

Step	Commissioning activity	What is done	Cold start-up steps	SOP that trains it	Other OTS procedure
CS-01	Pre-start-up safety review and walkdown	PSSR: punch list A closed, blinds removed, manways closed, head count, permits closed	—	SOP-SO3-01 Pre-start-up checks	—
CS-02	SIS validation (SAT) and ESD test	Every trip and permissive tested end to end; ESD push-button; trip response drilled with Operations	—	SOP-SO3-01 Pre-start-up checks; SOP-SO3-05 Emergency shutdown and trip response	Emergency shutdown: acid leak at the IPA pumps
CS-03	Acid circulation established at 98.5 %	T-301 and T-302 in circulation at design flow; spare pump switch-over demonstrated	P8	SOP-SO3-09 IPA acid pump switch-over P-301A → P-301B	—
CS-04	Molten sulfur on recirculation	Sulfur system hot, P-101A on recirculation, filter changeover demonstrated	P8	SOP-SO3-10 Sulfur pump and filter switch-over	—
CS-05	Blower on line, air through the train	K-101 running, FIC-101 in MAN at start-up air	P1	SOP-SO3-02 Cold start-up through the SIS	Cold start-up through the SIS
CS-06	Furnace and catalyst preheat	HV-112 open, H-101 fired, TIC-201 in AUTO on the WHB bypass, beds brought above strike, bed 1 inlet into its start window	P2 P3 P4 P5 P6	SOP-SO3-02 Cold start-up through the SIS	Cold start-up through the SIS
CS-07	Start-up line closed and locked	HV-112 closed and locked before sulfur	P7	SOP-SO3-02 Cold start-up through the SIS	Cold start-up through the SIS
CS-08	First sulfur through the SIS	Permissives met, sulfur admitted at low output, H-101 off once sulfur burns	P9 P10	SOP-SO3-02 Cold start-up through the SIS	Cold start-up through the SIS
CS-09	Loops to AUTO, ratio and cascade	TIC-201…251 AUTO; FIC-101 CAS on the ratio; AIC-104 AUTO with FIC-102 CAS	P11 P12	SOP-SO3-02 Cold start-up through the SIS	Cold start-up through the SIS
CS-10	Line-out and ramp to design rate	Sulfur raised in steps to design; SO_2 strength on its set point	P13	SOP-SO3-02 Cold start-up through the SIS; SOP-SO3-06 Production rate change; SOP-SO3-07 SO_2 strength change	Cold start-up through the SIS
CS-11	Initial operation and shift routine	Hold in band with stack SO_2 below the permit; shift hand-over and rounds started	P14	SOP-SO3-02 Cold start-up through the SIS; SOP-SO3-12 Shift hand-over and routine rounds	Cold start-up through the SIS
CS-12	Planned stop to hot standby and hot restart	Demonstrate the planned stop and the hot restart before the test (typical commissioning practice)	—	SOP-SO3-04 Normal shutdown to hot standby; SOP-SO3-03 Hot restart from hot standby	Planned shutdown (hold the catalyst hot); Hot restart from hot standby
CS-13	Fine tuning of the bed inlets	Loop tuning and bed-inlet adjustment by the method of the optimisation SOP, within the operating window	—	SOP-SO3-08 Stack SO_2 optimisation of the bed inlets	Optimisation: lowest stack SO2 at full rate (live, 45 min)
CS-14	Performance test (72 h)	Test against the guarantees below	—	SOP-SO3-12 Shift hand-over and routine rounds	—
CS-15	Handover and maintenance isolation	Handover certificate; first maintenance isolation under permit (LOTO) demonstrated	—	SOP-SO3-11 Isolation for maintenance (LOTO): economizer E-204	—

7.2 Cold start-up through the SIS (procedure steps as tracked by the simulator)

Step	Text
P1	Start K-101 and set about 80–120 kNm³/h of air with FIC-101 in MAN
P2	Open the start-up line HV-112 so hot gas also reaches beds 4 and 5
P3	Light H-101 and fire it at 50 % or more (HIC-109)
P4	TIC-201 in AUTO with its set point between 400 and 520 °C (hot gas through the WHB bypass)
P5	Beds preheated: TI-202 ≥ 400 °C and TI-232 ≥ 360 °C
P6	Bed 1 inlet brought to 400–440 °C (TIC-201 set point ≤ 440)
P7	Close and lock the start-up line HV-112

Step	Text
P8	Acid circulating in T-301 and T-302 and a sulfur pump running
P9	Admit sulfur through the SIS with FIC-103 in MAN at ≤ 25 %
P10	H-101 shut off once the sulfur burns (TI-106 rising)
P11	All bed-inlet loops TIC-201…TIC-251 in AUTO
P12	Air on ratio (FIC-101 CAS); SO2 strength on AIC-104 AUTO with FIC-102 CAS
P13	Line out: sulfur 40.9 ±0.5 t/h in AUTO, SO2 12.0 ±0.3 %
P14	Hold 10 minutes: stack SO2 ≤ 500 ppm, TI-203 < 640 °C, sulfur and SO2 on target

8 · Performance test

After a stable run at design rate the plant is tested for 72 h (typical) against the guarantees. Targets are the model design values (SO3-V1-PE-HMB-001); tolerances are assumed.

Tag	Quantity	Target (model design)	Unit	Acceptance criterion
FIC-103	Sulfur burnt	40.94	t/h	≥ design rate, 72 h average
FI-309	Acid production (100 % H_2SO_4)	125.0	t/h	≥ design, 72 h average
XI-308	Overall SO_2 conversion	99.80	%	≥ design, from stack SO_2 and bed-1 inlet SO_2
AIT-306	Stack SO_2	292.5	ppm	≤ design + 1 % analyzer allowance (assumed); never above the 500 ppm permit
FI-108	Steam export (4.0 MPa g)	94.0	t/h	≥ design − 2 % (tolerance assumed)
AIT-307	Stack acid mist	10.5	mg/Nm³	≤ design; alarm at 40
K-101	Blower power	4200	kW	≤ design + 3 % (tolerance assumed)
AI-104	SO_2 into bed 1 (test condition)	12.0	% vol	design ± 0.3 % (test condition, assumed)

8.1 Worked calculation · conversion from the stack analyzer

Gas leaving ÷ gas entering, from the model design point: $G = 10^6 \cdot y \cdot (1 - X)/ppm = 10^6 \times 0.120 \times (1 - 0.99800) / 292.5 = 0.820$.

Test averages at design: ppm = 292.5, y = 12.0 %. $X = 1 - ppm \cdot G/(10^6 \cdot y) = 1 - 292.5 \times 0.820 / (10^6 \times 0.120) = 99.80$ %. The 500 ppm permit corresponds to X = 99.66 %.

9 · Checks against the model and the other documents

Computed by the same function the tab runs (ecChecks), in node at generation; the tab recomputes them live.

#	Check	Found	Result
C1	Every SIS trip in the C&E (sf.json) has a SAT step with its set point and delay	6 trip rows, 6 covered	✓
C2	Every trip evaluated by the model logic (tripConditions) has a SAT step	6 model trips, all covered	✓
C3	Every start permissive (purgePermissives) has a SAT test	8 permissives, all covered	✓
C4	Every equipment item on PFD-201 (pf_spec.json) is in a procurement package	13 items, all in a package	✓
C5	Every SOP in the register (spMakeSops) maps to a commissioning step, and every SOP cited exists	12 SOPs, all mapped	✓
C6	Every step of the cold start-up procedure (PROCS.startup) is mapped to a commissioning step	14 steps, all mapped	✓
C7	Performance-test targets equal the model design values (SO.sum of snapDesign)	FIC-103 40.94 · FI-309 125 · XI-308 99.8 · AIT-306 292.5 · FI-108 94 · AIT-307 10.5 · K-101 4200 · AI-104 12	✓
C8	Catalyst in PK-04 equals the Design tab (dsCalc)	Design tab 187.1 t, package 187.1 t	✓
C9	Blower power in PK-01 equals the model (econ kW at design)	model 4200 kW, package 4200 kW	✓
C10	Schedule network: one unbroken critical path from award (K0) to handover (HO)	K0 → BED → LLP → F_CONV → ERC → PIP → EI → MC → PRE → DRY → CATL → COM → SU → RAMP → PT → HO · 34.00 months	✓
C11	Catalyst loading starts after the F-101/B-101 refractory dry-out (moisture kept off the catalyst)	dry-out ends month 31.00, loading starts month 31.00	✓

10 · Engineering queries

Query	Discipline	Subject	Finding	Proposed action
EQ-ec-01	Process / Mechanical	Acid coolers and sulfur filters have no tag or datasheet	PK-08 (acid coolers) and the sulfur filters of PK-09 cannot be requisitioned: PFD-201 does not show them, SO3-PID-002 refers to an acid-system P&ID (note 2) that is not issued, and the filters appear only through PDG-103.	Issue the acid-system P&ID; assign equipment tags (e.g. acid coolers on the IPA and FAT loops, sulfur filters A/B); add datasheets to the equipment list SO3-V1-ME-EQL-001 with the duties from an acid-loop heat balance.

Query	Discipline	Subject	Finding	Proposed action
EQ-ec-02	Instrumentation (SIS)	SIS architecture not fixed before the SIS purchase order	The SRS required SIL is not met by the 1oo1 architecture for 5 SIF(s): SIF-01 FSLL-100: required SIL 2 (PFD ≤ 1.6e-3), achieved SIL 1 (PFD 1.8e-2); SIF-02 TSHH-203: required SIL 2 (PFD ≤ 8.9e-3), achieved SIL 1 (PFD 1.3e-2); SIF-03 FSLL-301: required SIL 2 (PFD ≤ 3.3e-3), achieved SIL 1 (PFD 1.8e-2); SIF-04 FSLL-302: required SIL 3 (PFD ≤ 4.8e-4), achieved SIL 1 (PFD 1.8e-2); SIF-05 ASHH-306: required SIL 1 (PFD ≤ 1.7e-2), achieved SIL 1 (PFD 3.8e-2). PK-11 and the XV-103 valve in PK-10 cannot be bought or FAT-tested against a final C&E until this is closed.	Decide the architecture (e.g. a second shut-off valve XV-103B in 1oo2, redundant transmitters) before the PK-11 MR; re-verify the PFD; revise the C&E and the SAT steps.
EQ-ec-03	Mechanical / Process	Steam drum and BFW system outside the model	PK-03 needs the steam drum, blowdown and BFW pumps to raise 94 t/h at 4.0 MPa g, but the model and PFD-201 stop at B-101 / the economisers (BFW 150 t/h). Boil-out, steam blowing and the performance test of steam export depend on them.	Define the steam-system battery limit; add the drum and BFW equipment to the EQL and to PK-03 or a utility package; include them in the WHB & steam system of the MC checklist.
EQ-ec-04	Mechanical	Converter diameter is an assumption	The converter MR (PK-04) uses inside diameter 13.0 m, which the Design tab labels assumed; bed depths, grid loads and the catalyst loading plan follow from it.	Confirm the diameter in the R-201 datasheet (SO3-V1-ME-EQL-001) before the long-lead MR; record the bed depths for the catalyst loading procedure.
EQ-ec-05	Instrumentation / Operations	Proof-test interval versus turnaround interval	The SRS sets a 1-year proof test with a full stroke of XV-103; sulfuric-acid plants typically run 2–3 years between turnarounds, and XV-103 cannot be full-stroked with sulfur flowing without a bypass.	Specify partial-stroke testing or an online test bypass for XV-103 in PK-10, or align the proof-test interval with the turnaround and re-verify the PFD.

11 · Assumptions

- Lead times, inspection levels, schedule durations, the 72 h test length and the acceptance tolerances are typical practice, assumed for teaching.
- Converter inside diameter 13.0 m is the Design tab assumption.
- Acid coolers, sulfur filters, steam drum and BFW pumps are outside the model; they are listed as packages or scope with no equipment tag (see the queries).
- SAT methods are typical; set points, delays and effects come from the C&E sheet and the model logic.
- No client, contractor or vendor is named or represented.